Unravelling the mysteries behind orcein dye, Part B: from lichen-based synthesis to molecular degradation.

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Abstract

When mixed in ammonia solution, certain lichen species can produce a red to purple hue colourant called orchil, used as a dye in historical artefacts. In the case of the study of Mont-Saint-Michel's manuscripts, this dye constituted by a series of orcein molecules, might have been produced with lichen species such as *Roccella fuciformis* or *Roccella phycopsis*, both locally found in the area. The objective of this paper is to evaluate the impact of synthesis parameters on lichen-dye orchil synthesis colour, such as lichen quantities, ammonia and O_2 ratios, but also lichen's nature on the produced colour, and complements an initial approach using orcinol as a precursor.

Purple precipitate apparition, only observed with *R. fuciformis*-based synthesis, allowed experimental separation of purple and red-orange molecules by sample filtration. This key experiment confirmed the results of the PLS analysis performed on the same colourant, but synthesised with pure orcinol molecules, which suggested that β/γ-orcein molecules were majorly responsible for the apparition of purple hues. Using *R. phycopsis* as a starting lichen species led to variations in both colour development and molecular signatures, as investigated by UV-Vis spectroscopy and HPLC-DAD-MS. Higher molecular production was achieved, with a darker solution compared to *R. fuciformis*, alongside with higher production of β/γ-aminoorcein.

Finally, artificial ageing experiments confirmed a higher degradation rate of the β/γ-orcein compared to α-orcein, resulting in significant colour changes in purple parchments.

Keywords

Orchil; lichen; synthesis; chromatography; molecular composition; colour assessment; ageing

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1. Introduction

Recent material analyses carried out on manuscripts produced at the Mont Saint-Michel Abbey scriptorium between 950 and 1100 have revealed the important use of orchil in decorative and illustrative work. Obtained by mixing colourless lichen and stale urine[1–9], orchil dye was identified in these manuscripts, exhibiting various colour shades, going from red-brown to deep purple colours. Variation in orchil colour is a known phenomenon[10,11], as orchil was used as one of the first pH indicators, named litmus paper. Furthermore, the pioneer works of Beecken et al. [12] helped identify most compounds in orchil molecular composition and assess the colour of each isolated main molecule.

Our recent study on the impact of orcinol-based orchil dye production participated in identifying the role of orcinol concentration, O_2 and ammonia ratios on the observed colour [**Paper A**]. This investigation revealed that orcein molecules composed of three orcinol groups (named α-orcein) induced a light red-orange colour, while more complex orcein families, composed of 4, or more orcinol groups (namely β-, γ-, δ-, ζ- or even dimers of α-orcein), generally brought darker purple shades. To produce these molecules, a minimal orcinol concentration was required, alongside optimal O_2/ammonia ratios. Changing one parameter impacts the molecular production, and thus, the resulting colour.

The first part of this article consists in investigating the impact of lichen quantity, ammonia and O_2 ratios on lichen-based orchil dye colour. Thanks to the depside presence in their chemical composition (i.e., erythrin and lecanoric acid) and their local presence near the Mont Saint-Michel Abbey, the *Roccella fuciformis* and *Roccella phycopsis* lichen species were selected[13]. Since this experiment required preliminary tests and a high amount of lichen, it was decided to use *R. fuciformis* as the main lichen. Furthermore, due to its high natural abundance near the Mont Saint-Michel, it can be hypothesised that colourants found in the manuscripts were produced locally with this species[13,14]. Based on an experimental protocol developed in our laboratory for orcinol-based orchil synthesis, the five experiments were conducted with *R. fuciformis* as precursors. HPLC-DAD-MS and transmittance analysis will be performed on the obtained solutions throughout the 43-day synthesis, managing colour and chemical composition evolutions.

After producing red-orange and purple-coloured solutions, it was decided to investigate further the purple-yielding synthesis. After two weeks, a purple precipitate was formed on the bottom of the flask, with a red-orange solution, indicating possible separation of the coloured molecules during the synthesis process. After 40 days, both the liquid solution and the precipitate exhibited the same hues. This initial series of experiments enabled us to determine the parameters required to obtain a purple dye. A second series with these conditions was prepared to test the reproducibility and enable more accurate monitoring by analysing the liquid and solid components obtained during the synthesis of this dye. In this part of the study, synthesis was carried out with the same lichen species but collected 25 years ago.

As previously mentioned, *R. phycopsis* lichen species can also be collected near the Mont Saint-Michel and is present on walls of the abbey (personal observation of J. Boustie). Monitoring orchil dye production with this lichen was also performed in experimental conditions identical to those which rendered purple hues with *R. fuciformis*. Interspecies variations in kinetics of colour appearance and chemical composition were monitored as above.

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The last part of this study describes artificial ageing of orchil dye. Sheepskin parchments were prepared and coloured with the obtained orchil dyes. A gradual artificial ageing process was carried out to follow the molecular degradation of all detected orcein molecules.

2. Material and method

2.1. Chemicals:

All chemicals used in this paper are identical to the ones used in Paper A. The names and abbreviations used in this paper are all explained in the previous study (see Paper A).

Lichens were collected by J. Boustie (University of Rennes) in Brittany (North-West France). *Roccella fuciformis* used to study the impact of synthesis parameters was collected in 2024 on seashore rocks (voucher: JB/24/300), while for the comparative experiment, the lichen was collected in 2005 (voucher: JB/05/71). *Roccella phycopsis* was collected also in 2005 (JB/05/72), near Saint-Malo. Both lichens were selected, due to their natural presence near the Mont Saint-Michel, for our study on orchil dye origins.

2.2. Orcein dye analysis

2.2.1. Lichen-based synthesis

Preliminary tests were carried out using *R. fuciformis* as a dye source, allowing determination of the lower and upper quantities of lichen, determined regarding their capacity to produce red-orange- or purple-coloured solutions. **Table 1** presents the five experiments carried out to study the impact of lichen quantity, ammonia and O_2 rates on orchil dye chemical composition and colour.

The 1% and 12.5% (v/v) ammonia solutions were prepared by diluting the 25 %(v/v) commercial NH_4OH solution by a factor of 25 and 2 in ultrapure water. Lichen was mixed with ammonia solutions, left to react for about a month and a half and agitated twice a day. HPLC-DAD-MS and transmittance analysis took place on the following days: 0, 1, 2, 4, 7, 8, 9, 14, 16, 18, 35, 39 and 43. Presented in Figure SI1 (Supplementary), the experimentations using lichen required higher reactor volume compared to orcinol-based syntheses (paper A), thus higher O_2 presence in the headspace of the sealed vessel experiments. The sealed vessels (50 mL vessels with a screw cap) will be referred to as hypoxic environments and opened vessels as an oxic environment.

2.2.2. Artificial ageing protocol

Sample preparation, ageing, and solid–liquid extraction protocols were identical to those described in **Paper A**.

With light being considered as the most impactful degradation parameter, ageing under light exposure was carried out on an ALPHA+ Xenotest chamber (Atlas). The samples were placed on a rack rotating around a xenon lamp equipped with a filter (Xenochrome 320) simulating the solar spectrum behind 3 mm thick glass (indoor conditions). The ageing parameters were as follows: the chamber temperature was 35.5°C (± 1.4°C), the black-standard temperature was 55.6°C (± 1.5°C), the irradiance level was set to 50 W/m² (± 1.5 W/m²) and the relative humidity was 50% (± 0.6). Samples were aged up to 189 hours, cut in 10-11 small bands (1.5 cm long), and placed in the dark in a sealed bag until dye characterisation by reflectance spectroscopy and HPLC-DAD-MS.

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2.3. Analytical instruments and analytical conditions

2.3.1. HPLC-DAD-MS

HPLC analysis was carried out on an UltiMate 3000 chromatograph, equipped with an HPG 3400 RS quaternary pump, WPS 3000 TRS autosampler, TCC 3000 SD column compartment, and a DAD 3000 RS diode array detector, driven by Xcalibur and Chromeleon Xpress softwares, all from ThermoScientific (Waltham, MA, USA).

The mobile phase consisted of an acetonitrile/water-based gradient, with channel (A) being pure water, channel (B) pure acetonitrile and channel (C) a 10% (v/v) formic acid solution of pH = 1.45, measured with Seven Easy pH meter from Mettler Toledo (Columbus, OH, USA). The gradient went from 75/15/10 to 40/50/10 in 40 minutes, followed by a two-minute isocratic segment at 40/50/10. Afterwards, eluent composition was set at initial conditions of 75/15/10 for 18 minutes of system stabilisation. Column temperature was maintained at 30°C, and flow rate was set at 0.2 $mL.min^{-1}$.

The separation was achieved using the Uptisphere Strategy C18HQ stationary phase of 5µm in 125 x 2.1 mm column (Advion Interchim Scientific, Montluçon, France).

Two detectors, DAD and MS were coupled in series. The DAD operated in full scan mode over the 200–800 nm wavelength range. To selectively monitor the coloured compounds of interest, chromatograms were extracted at 274 nm, 410 nm, 485 nm and 560 nm. Mass spectrometric detection was performed using an electrospray ionisation (ESI) source in positive mode. The mass range was set from m/z 70 to 1200. Experimental parameters included a spray voltage of 3.5 kV, a capillary temperature of 100°C, and sheath and auxiliary gas flow rates of 10 and 30 arbitrary units, respectively. The injection volume was 5 µl.

2.3.2. UV-visible spectroscopy

UV-visible spectroscopy analysis (transmittance and reflectance spectra) was recorded from 200 to 800 nm with 1 nm resolution using a UV-Vis-NIR Cary 5000 spectrophotometer (Agilent Technologies, Santa Clara, USA). Transmittance and reflectance experiments were carried out to measure sample colour evolution through the determination of L*a*b* coordinates, expressed in the 1976 CIE convention, with a D65 illuminating source and a 2° observation angle[15]. *L** represents lightness (scale from 0 to 100), *a** the green-red axis, and *b** the blue-yellow axis (both scales going from -128 to +128).

Transmittance measurements were done to follow the kinetics of synthesis reactions in various experimental conditions. Analyses were carried out in a 1 mm wide cuvette. To avoid signal saturation, samples can be diluted by a factor of 10 in pure water. In an oxic environment, strong solvent evaporation prevented analysis after 9 days of synthesis. The remaining experiments were monitored over 43 days as follows: after days 0, 1, 2, 4, 7, 8, 9, 41, 16, 18, 28, 35, 39 and 43.

Reflectance spectra were measured on parchment samples painted with the orchil dye obtained after 43 days of reaction. The objective of this analysis was to measure colour evolution of orchil dyes applied on parchment during artificial ageing. Spectra were recorded using an integrating sphere (polytetrafluoroethylene (PTFE) coating) with an inserted gloss trap to exclude the specular reflectance component. The spot size was 1 cm x 0.5 cm, and the spectral resolution was 1 nm. A PTFE white standard (Spectralon® Reflectance Standards, Labsphere Inc.) was used for baseline correction before reflectance measurement.

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3. Results and discussion

3.1. Orchil synthesis

3.1.1. Characterising orchil composition and colour for different synthesis from Roccella fuciformis

The study of the experimental synthesis parameters presented in **Table 1**, was determined with orcinol-based orchil synthesis experiments, presented in **Paper A**.

Error! Reference source not found. presents the evolution of the dye synthesis in Lf_001 and Lf_002 over time, where the parameter tested is the lichen quantity (0.1 g for Lf_001 and 5 g for Lf_002). For HPLC-DAD-MS analysis only 3 chromatograms are presented here, on days 4, 18 and 43. L*a*b* coordinates evolution during both syntheses were traced with all measurement points (**Error! Reference source not found.B** and **1D**). After 18 days of synthesis, it was decided to separate the lichen from the solution, as previous experiments showed faster colour evolution once the two parts were separated (preliminary experiments and results not shown). Analyses performed after day 18 are presented in green areas of diagrams C and D. All experiments presented in this figure correspond to the analysis of the filtered synthesis solution at the indicated time. Analysis of the precipitate will be presented in Section 3.1.2.

Only the results of experiments Lf_001, Lf_002 and Lf_005 are presented, because the Lf_003 and Lf_004 experiments were stopped on the 9th day, as discussed above.

Thanks to HPLC-DAD-MS and transmittance monitoring of the three successful syntheses, a strong resemblance between Lf_001 and Lf_005 in terms of colour and chemical composition over time was observed (see **Error! Reference source not found.A** and **Figure SI2**). A red-orange solution colour was obtained at the end of the experiment, with an extremely low presence of compounds from the α-orcein family. On the contrary, Lf_002 exhibits a deep purple solution colour, with an important variety in orcein molecules produced, especially after the removal of the lichen from the solution. As displayed in **Error! Reference source not found.C**, mainly α-HO (30.6 min) was produced before the lichen removal, while the solution turned to contain mostly β/γ-AO and β/γ-HO after. These observations are in perfect agreement with L*a*b* measurements and previous conclusions on experiments conducted with pure orcinol as an orcein precursor [**Paper A**]. Before separation, both *a** and *b** values were positive (implying a red-orange hue), because the dye was mostly composed of α-orceins. The separation induced important drop in *b** scale, due to predominant pathway of β/γ-orceins synthesis.

As presented in **Figure 2**, this experiment allowed selecting synthesis conditions to produce purple hues with *R. fuciformis*. Similarly to orcinol-based synthesis, a high lichen-to-reaction-solution ratio, low ammonia content and presence of O_2 are required. The detection of purple precipitate, only observed in Lf_002 synthesis, raised the question regarding the possibility of isolating purple molecules from the red-orange ones. The next section will be discussing this matter.

3.1.2. Isolation of the purple precipitate during Roccella fuciformis orcein-based synthesis

As presented in **Figure 2**, mixing *R. fuciformis* and ammonia solution generates a milky-white suspension that precipitates when left untouched. After 43 days of synthesis, both Lf_001 and Lf_005 synthesis exhibited a red-orange coloured solution, with grey particles on the bottom of the reaction vessel (see **Figure SI3A**), while Lf_002, the only experiment leading to a purple-coloured solution, had the precipitate exhibiting the same hues as the solution (see **Figure SI3B**).

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As seen during the Lf_002 experiment, after separation of the lichen and synthesis solution, the precipitate already exhibited the desired purple shades, while the solution colour was red orange. It was hypothesised that physical isolation of the particles from the solution, followed by a solubilisation step, can help identify the molecules responsible for those colours. The Lf_002 experiment was reproduced and named Lf_002_1.

It is important to note that this new synthesis used *R. fuciformis* as a precursor, but due to a lack of sample, another batch of lichen had to be used. Lf_002_1 lichen was collected in the same region but 20 years prior to the batch used in previous experiments, leading to potential variations in its chemical composition. Regardless of the lichen batch variation, the purple precipitate appeared 10 days after the beginning of the synthesis, and lichen was manually extracted from the solution five days afterwards. The solution/precipitate vial is presented on the left of **Error! Reference source not found.A**. The Lf_002_1 solution was filtered through a 0.2 µm PTFE syringe filter to successfully separate the solution from the precipitate. The orange solution was directly analysed by HPLC-DAD-MS, and the chromatogram is presented with the black trace on **Error! Reference source not found.B**. Solubilisation tests were carried out on the purple precipitate. The solvents used for this experiment were pure water, 100% methanol, and the synthesis solution (1% NH_3). After ten minutes, all solutions were filtered, as presented on the right of **Error! Reference source not found.A**. Corresponding HPLC-DAD-MS analyses are displayed in **Error! Reference source not found.C**.

The successful separation and analysis of the two coloured phases after 15 days lead to experimental determination of the role of each orcein molecule family. The red-orange solution (liquid phase) presented two molecules of the α-orcein family (α-AO and α-HO). The purple solution, obtained with the solubilisation of the purple precipitate with methanol or 1% ammonia solution, showed a greater chemical diversity, with the presence of β- and γ-orcein molecules. These results match perfectly the PLS analysis performed previously [ref. **paper A**], where it was concluded that α-orcein molecules brought light red-orange hues, while β/γ-orcein molecules brought deep purple colours.

The unfiltered Lf_002_1 solution was then left for 25 days before being analysed again. As presented in **Figure SI4A**, after 40 days of synthesis, both the solution and the precipitate expressed purple shades. It can be estimated that the purple molecules produced near the precipitate were slowly solubilised, thanks to the presence of ammonia at 1% (v/v).

To verify this hypothesis, Lf_002_1 was filtered after 40 days of synthesis and characterised through HPLC-DAD-MS (see the blue upper trace of **Error! Reference source not found.B**). It can be noted that the main molecule at 15 days (α-AO) remains highly detected, but the chemical signature at 40 days is far more complex than the previous one. Molecules belonging to the β/γ/δ/ζ-orcein families, presented previously in **Paper A**, partially identified in the precipitate, are now solubilised in the liquid phase. The apparition of δ/ζ-orcein molecules can be noted, but their lower relative proportion seems to indicate later synthesis of these compounds compared to β/γ-orceins. Thanks to these results, we can estimate a local formation of these molecules, near the precipitate area, followed by a slow solubilisation, thanks to the presence of ammonia in solution, in the case of *R. fuciformis*-based orchil synthesis.

This experiment provided several significant insights. It allowed experimental identification of the role of each molecule in the rendered colour, in perfect agreement with PLS analysis performed with orcinol-based experiments. Slow solubilisation of the purple molecules was observed with the two experiments, with purple colouration in the liquid phase. Variations in the chemical composition of both samples were noted throughout the experiment. Low α-AO amounts were produced with the 2024 lichen batch (see **Figure 1C**), while for the 2004 batch, this molecule has the higher peak relative intensity (**Error! Reference source not found.B**). It is also noted that the β/γ-HO production is very

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different between the two batches. With fresh lichen, β-HO and γ-HO were the most intense peaks at day 43, while only one of them was produced with the older lichen batch. These differences may stem from intraspecies variation resulting from subtle changes in their chemical composition. Factors such as their location, environmental conditions, or degree of freshness could also account for these compositional variations. A probable small-sample bias occurred in this experiment, as only 5 grams of lichen were used. Historically, recipes describe much larger batch sizes, from about half a kilogram[2] up to 100 kg[16], which would likely reduce the small variations observed here. Nevertheless, despite the 20-year difference between the two lichen samples used, both specimenswere able to produce the expected deep purple colour after 40 days.

3.1.3. Synthesis of orchil from Roccella phycopsis

In this part of the study, orchil synthesis will be carried out using *R. phycopsis* in the experimental conditions that led to purple hues. The synthesis will be called Lp_002. As demonstrated in Lf experiments and with orcinol-based orchil synthesis, variations in synthesis parameters can strongly impact the detected colour. Thus, the Lp_002 experiment was carried out concurrently to the Lf_002, keeping the same environmental conditions (light exposure or variation of temperature). The evolution of its chemical composition and colour are presented in **Figures SI5A** and **SI5B**. Variation induced by lichen species is presented with the chromatograms in **Error! Reference source not found.A** and **4B**, respectively, at days 18 and 43. *R. fuciformis* is represented with the blue trace, while the red trace represents the *R. phycopsis* analysis.

The main visible differences observed between Lp_002 and Lf_002 were the kinetics in colour apparition and the absence of particles with the newly tested lichen. It is interesting to note that observation of the orchil-based paint layers in the Mont Saint Michel manuscripts sometimes reveals very different textures, sometimes with a very thick appearance and sometimes very transparent, like watercolours. Based on the results of this study, this appearance could be correlated with the presence or absence of a precipitate during the dye synthesis process. The red-orange solution appeared within minutes after mixing the 1% v/v ammonia solution with *R. phycopsis*, without cloudiness (thus no particles in suspension). Throughout the experiment, the observed colour was systematically darker than the one of *R. fuciformis*.

Error! Reference source not found.A presents the HPLC-DAD trace recorded at 560 nm, after 18 days of synthesis, before separation of the lichen from the synthesis solution. A more complex chemical composition with higher peak intensity is measured with *R. phycopsis* (upper trace) when comparing it to *R. fuciformis* synthesis (lower trace). Molecules such as β/γ-AO (tr = 18.6 min), α-AO (tr = 21.2 min) and α-HO (tr = 30.2 min) were detected in both experiments with different relative intensitys. But the presence of β/γ-AO at 24.2 minutes is exclusively noted in Lp_002. Higher peak intensity and more complex chemical composition can explain the variations in the observed colour, where a deeper red-orange hue was achieved with *R. phycopsis* after 18 days.

Error! Reference source not found.B presents the HPLC traces of both lichens after 43 days of synthesis. The lichen species impact on the chemical composition of the dye remains observable. With *R. phycopsis* synthesis, α-AO molecule (21.2 minutes) detection is far greater than with the other lichen. Plus, the relative peak intensity of the β/γ-HO and β/γ-AO peaks (respective tr of 23.4 and 24.2 minutes) are different in both experiments. The *R. fuciformis* synthesis produced little or no β/γ-AO, compared to the *R. phycopsis*, after 40 days.

It is important to note that α-AO variation, observed between two *R. fuciformis* experiments (Lf_002 and Lf_002_1), can be explained by potential intraspecies variations or the impact of the lichen's freshness on the final orchil chemical signature.

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The same conclusion can be drawn with the interspecies variation study between the Lf_002 and Lp_002 experiments, since both lichens were also collected 20 years apart. But differences between *R. phycopsis* and *R. fuciformis* (from the 2 batches) were systematically observed, inducing clear interspecies variations in orchil dye production. Visual clues, such as a slower colour apparition and milky-white haze solution in the case of *R. fuciformis* syntheses, are attributed to interspecies variations, and in terms of final solution colour, a brighter dye was obtained in the case of Lf_002 compared to Lp_002. While both exhibited an almost identical chemical composition after 43 days, important differences were noted in terms of relative presence. *R. phycopsis* produced higher quantities of β/γ-AO, especially the molecule detected at 24 minutes. This difference could be a clue leading to lichen identification in dyes from historical samples. Because of the important intraspecies variations observed for *R. fuciformis* in this study (which can certainly also occur for other species), we must be cautious about identifying lichens based on the presence of specific coloured molecules. The specific chemical signatures are difficult to identify.

3.2. Artificial ageing

The evolution of the dye's colour and chemical composition upon artificial ageing was investigated using four samples: Lf_001 and Lf_005, both exhibiting a red-orange hue (set I), and Lf_002 and Lp_002, characterised by a deep-purple colour (set II). The dye solutions were directly applied to the prepared sheepskin, following the procedure described in the Material and Method section of **Paper A**.

After being painted, the mock-ups were left to dry for 48 to 72 hours before placing them in the ageing chamber. The ageing process was carried out in a stepwise manner to investigate both the degradation kinetics of the molecules and the colour evolution. The ageing process was regularly paused to collect small portions of the samples, used for UV-visible reflectance measurements, and HPLC-DAD-MS analysis. Mock-ups were aged for a total of 189 hours (with the exception of Lf_001, being aged for 165 hours, due to smaller sample size), with a total of 11 sub-samples to analyse.

Monitoring chemical composition through HPLC-DAD-MS analysis required a solid/liquid extraction protocol. Based on the literature[39-42] and previous experiments (**Paper A**), the water/methanol (1:1) solvent was found to be the best option.

Colour evolution of all four samples is presented in **Error! Reference source not found.**. **Error! Reference source not found.A** presents the visual evolution of all experiments throughout 189 hours of accelerated ageing; **Error! Reference source not found.B**, **5C** and **5D** respectively display the evolution of *L**, *a** and *b** parameters. **Error! Reference source not found.A** and **6B** respectively represent the evolution of the chemical composition of the set I and set II experiments, detected at 560 nm.

Thanks to L*a*b* measurements, it can be noted that in the *L**, *a** and *b** evolutions, distinctive patterns can be detected between the two sets of samples.

To analyse the results presented in **Error! Reference source not found.**, we need to start observations from a macroscopic point of view. Colour degradation, presented in **Error! Reference source not found.A**, displays important colour loss in set II samples, going from a light (lf_002) or deep (lp_002) purple-coloured sheepskin to a transparent red, while set I samples already exhibited a light red-orange hue at the beginning, exhibiting less evolution at the end than previously mentioned samples.

This colour evolution can be linked to the sample chemical composition presented in **Error! Reference source not found.**. In the case of set I samples, at t_0, only one orcein molecule was extracted from the

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parchment (α-AO), with very weak peak intensity, while 9 different molecules were extracted for set II samples (α-, β-, γ-, δ- and ζ-orcein). This variation explains the differences in observed colour at t_0.

During ageing, the colour evolution can be linked to molecular degradation, measured by HPLC and monitored in the case of the Lp_002 experiment (see **Error! Reference source not found.C**). Thanks to molecular degradation monitoring, it can be seen that more complex molecules (i.e., β-, γ-, δ- and ζ-orceins) tend to degrade faster than the α-orcein molecule detected at 21.2 minutes. Due to their higher sensitivity, *L**, *a** and *b** evolutions tracked by reflectance measurements presented in **Error! Reference source not found.B**, **5C** and **5D** show more variation in set II (purple lines) samples than in set I (orange lines). One interesting evolution regards the monitoring of set II samples on the *b** scale. A strong increase in *b** value is observed for the two samples between t_0 and t_{50} (Lp_002) or t_{79} (Lf_002), before a slow and steady decrease in *b** value until the last analysis. This exhibits a loss of blue shade in these samples during the first hours of accelerated ageing. Knowing their role in blue/red hues (see PLS results in **Paper A**), the loss of β-, γ-, δ- and ζ-orcein molecules explains this peculiar colour evolution.

4. Conclusions

The study investigated the impact of the lichen species and conservation conditions on orchil synthesis. To mimic the historical dye observed on the Mont Saint Michel's manuscripts, two lichens were selected: *Roccella fuciformis* and *Roccella phycopsis*, as both species are geographically present in the region and are considered plausible candidates for use in the historical dyeing process.

To study the impact of ammonia and O_2 rates, identified as key parameters in orcinol-based orchil dye synthesis, *R. fuciformis* species was selected due to its natural abundance. The conclusions regarding the role of each studied parameter are not changed from the experiences carried out with orcinol. Low quantities of lichen have systematically led to light red-orange-coloured solutions, the same as with high ammonia rates in sealed reaction vessels. Conditions to produce purple hues were reached with high lichen quantities, with a controlled O_2 rate and low NH_3 quantity. The use of a natural product as a reactant induced several changes in synthesis preparation, such as the lichen/reaction volume ratio. The 5 g of lichen were mixed with 15 mL of 1% (v/v) ammonia only, leading to a high presence of O_2, even in sealed large vessels. This, coupled with natural solution evaporation, led to unusable experiences.

Variations in orchil chemical signature observed between the Lf_002 and Lf_002_1 syntheses can be explained by both intraspecies variations and/or the impact of the lichen's freshness, due to a 20-years-older *R. fuciformis* lichen in the reproduced experiment. After 15-18 days, both solutions exhibited a red-orange hue, with purple-coloured precipitate. The liquid phase, analysed by HPLC-DAD-MS, showed no differences in terms of molecular composition, but important differences in relative proportions. Both solutions were composed of α-orcein, but the repeated experiment showed an α-AO peak far greater than the one in the experiment with recently collected lichen. After 40 days of synthesis, while both experiments exhibited a similar purple shade, their respective molecular compositions were different. This time, both β/γ-OH, the most intense peak in the first synthesis, were practically not observed in the repeated experiment. It was concluded that the presence of β/γ-AO in the second experiment could explain the apparent purple colour.

Interspecies variations were also investigated with *R. phycopsis* used as a lichen source in experimental conditions rendering purple hues with *R. fuciformis*. Changing the lichen's species induced several changes. First, colour appeared at a faster rate, with deeper intensity. As demonstrated by the HPLC-DAD analysis and comparison at days 18 and 43 of both syntheses, global intensities are far greater

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with *R. phycopsis* than with *R. fuciformis*. While the same mass of lichen was used in both experiments, this difference can be explained by the variety of depsides in both species, inducing different reaction kinetics. In the case of *R. phycopsis* experiment, no precipitate was observed, and purple-coloured molecules were directly produced in solution, as demonstrated by a greater presence of β/γ-orcein in the Lp_002 liquid phase analysis. After 43 days of synthesis, a deeper colour was reached with the *R. phycopsis* lichen species, compared to the *R. fuciformis* experiment in the same experimental conditions. While the overall orchil dye chemical composition remains similar regardless of the lichen species used, important variations were noted in terms of proportion. While β/γ-HO is one of the most intense peaks in both experiences after 43 days of synthesis, a far greater β/γ-AO production was noted in *R. phycopsis* than in *R. fuciformis* orchil dye analysis.

Conclusions drawn after the accelerated ageing protocol carried out with red-orange and purple orchil dye hues did not differ from the ones obtained with orcinol-based orchil dye synthesis. Purple samples exhibited greater changes in colour and chemical composition, with faster degradation of β-, γ-, δ- and ζ-orceins and the residual α-AO peak remaining still detectable after 189 hours of accelerated ageing. So, when interpreting analytical results from historical artefacts, particular attention must be paid to the dye's exposure to degradation factors and its state of conservation. These parameters directly influence the dye composition and its overall analytical response.

With this study, we have found out that lichen's nature, and potentially its freshness, impacts the final colour. In the historical recipes, the freshness is never mentioned, and it is possible that several lichen species were mixed during the synthesis, rendering identification of the lichen even more challenging. Furthermore, it is known that historical dyes were applied on the parchment, using various binders that can affect the rendered colour and its preservation. The impact of the binders on those aspects remains to be investigated, given the versatile nature of the chemical constituent of orchil dye.

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Author contributions: CRediT

Corresponding authors are required to acknowledge co-author contributions using CRediT (Contributor Roles Taxonomy) roles:

- Conceptualization – AM, SH, WN
- Data curation – SR, CD
- Formal analysis - SR
- Funding acquisition – AM, SH, WN
- Investigation – SR, CD
- Methodology - AM, SH, AT, WN
- Project administration – AM, SH, WN
- Resources - JB
- Supervision – AM, SH, WN
- Validation - SR
- Visualization – SR, CD

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- Writing – original draft – SR, CD
- Writing – review and editing – AM, SH, JB, AT, WN

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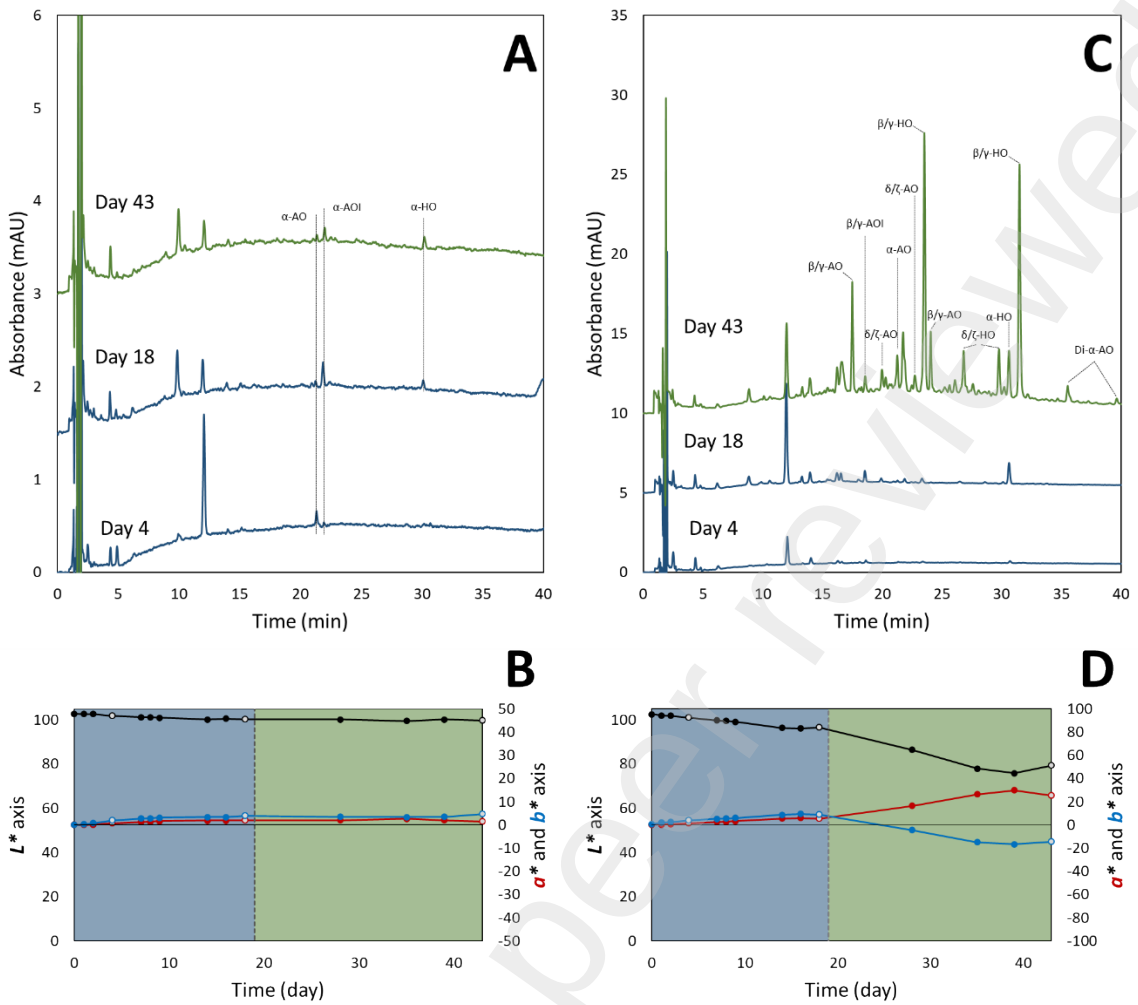

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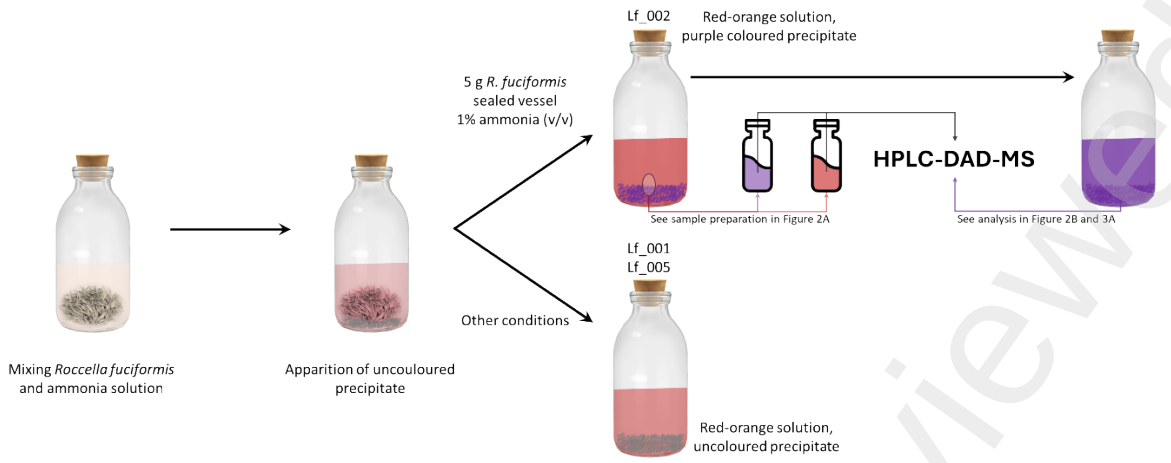

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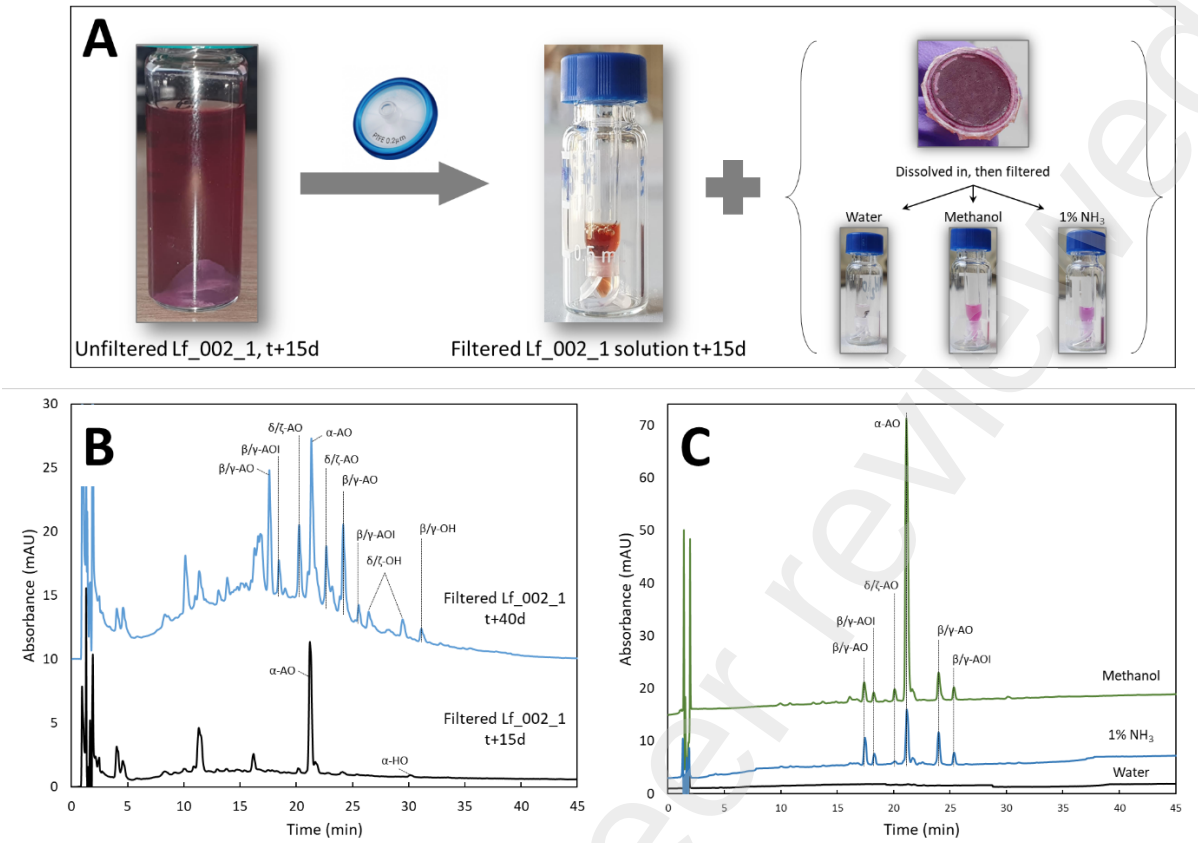

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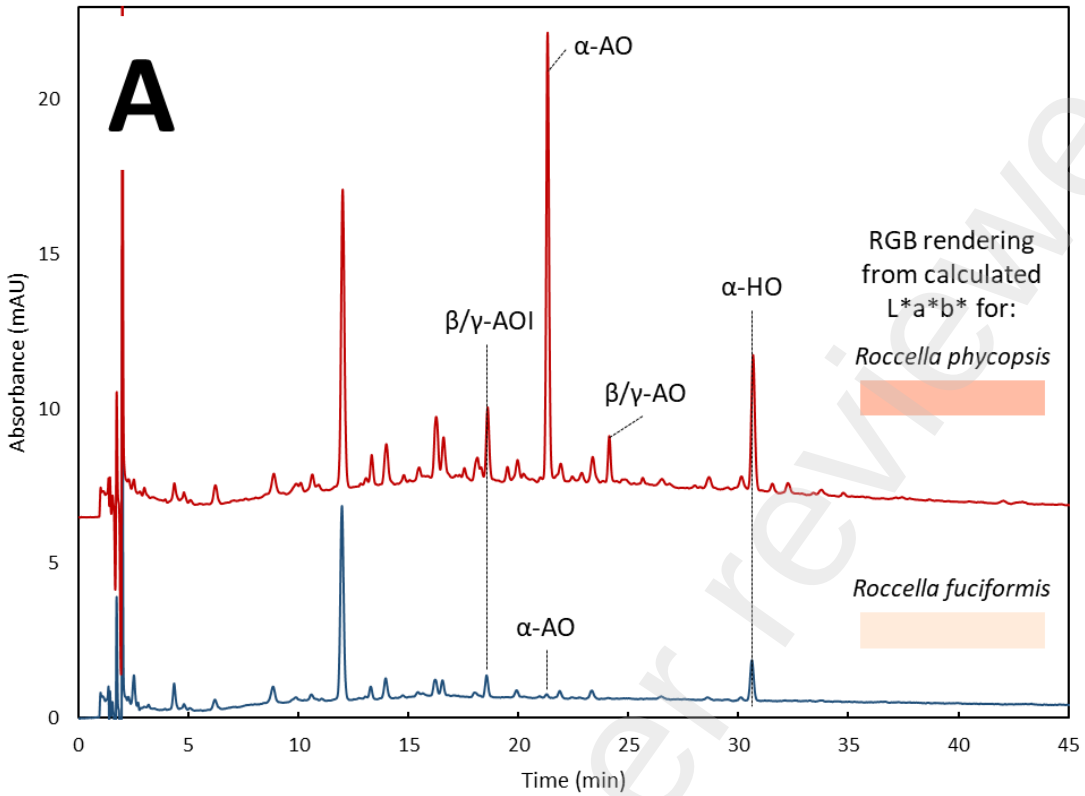

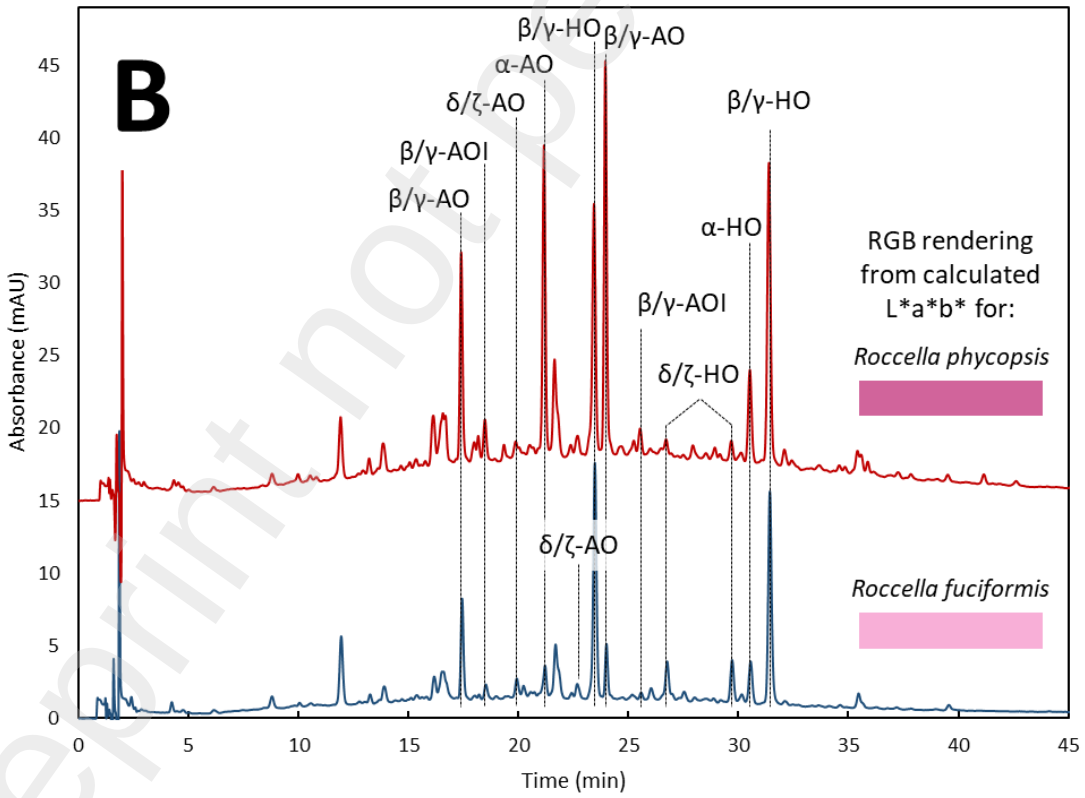

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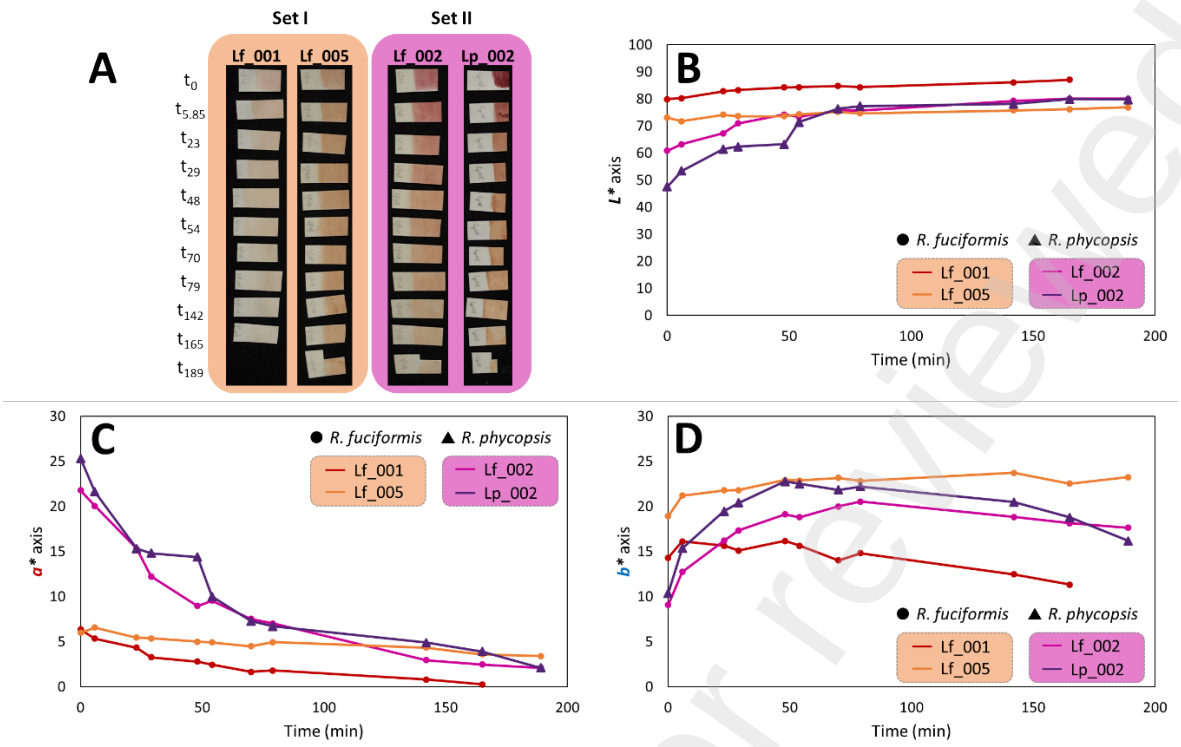

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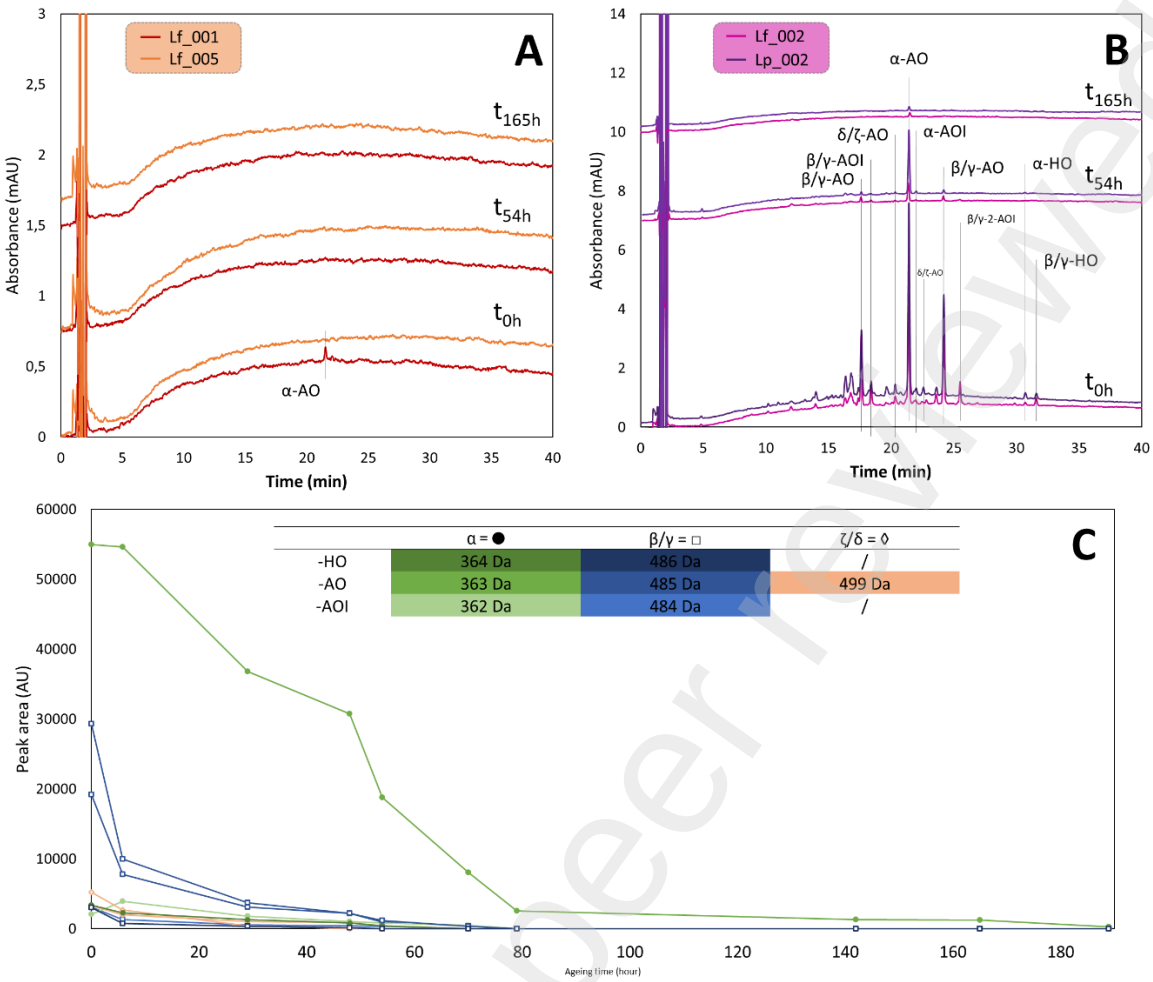

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***Figure 1**: Chromatographic monitoring at 560 nm (**A**, **C**) and colour evolution in the CIELAB space (**B**, **D**) during the synthesis experiments Lf_001(**A**, **B**) and Lf_002 (**C**, **D**). White dots in L*a*b* monitoring (**C**, **D**) represent the respective HPLC-DAD analysis displayed (**A**, **C**). HPLC-DAD-MS and L*a*b* colour monitoring experimental conditions are described in Material and Methods part 2.*

***Figure 2**: Representation of synthesis evolution and sampling scheme.*

***Figure 3**: Schematic representation of the experimental workflow (**A**) chromatographic profiles (560 nm) of the filtrates at 18 days and t+40 days (**B**), and chromatographic profiles (560 nm) of residue solubilization tests in water, 1% NH_3 and 100% methanol (**C**). HPLC-DAD-MS experimental protocol is described in Material and Methods part 2.*

***Figure 4**: Comparison of HPLC-DAD-MS analysis and obtained color representations between R. fuciformis (blue traces) and R. phycopsis (red traces) after 18 days (**A**) and 43 days (**B**) of synthesis. Synthesis and HPLC-DAD-MS protocols are detailed in Material and Methods part 2.*

***Figure 5**: Artificial ageing of 4 parchment samples (**A**), and respective L*, a* and b* monitoring through reflectance measurement (**B**, **C** and **D**). Artificial ageing and L*a*b* measurement analysis protocols are described in Material and Methods part 2.*

***Figure 6**: HPLC traces (560 nm) of experiment Lf_001 and Lf_005 (**A**), and Lf_002 and Lp_002 (**B**), at t_0, t_{54} and t_{165} and molecular peak area monitoring of Lp_002 experiment (**C**). HPLC-DAD-MS analysis and solid/liquid extraction protocols are described in Material and Methods part 2.*

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Name of the synthesis experiment	Lichen quantities[a] (g)	Ammonia concentration[b] (%)	Oxic/Hypoxic
Lf_01	0.1	1	Hypoxic
Lf_02	5	1	Hypoxic
Lf_03	5	1	Oxic
Lf_04	5	12.5	Oxic
Lf_05	5	12.5	Hypoxic

[a] Determined by preliminary experiments, not discussed in this paper

[b] 15 mL of solutions added to lichen

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Table 1: *Roccella fuciformis* orchil dye synthesis parameters

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